SMART INDIA HACKATHON 2026

Safety System for Low-Visibility Mining Operations

- **Problem Statement ID - SIH26007**
- **Problem Statement Title- Safe and Efficient Operation of Mine Vehicles in Fog and Low-Visibility Conditions in Open Cast Iron Ore Mines.**
- **Theme- Smart Automation**
- **PS Category- Hardware**
- **Team ID- SIH26-A0H-T322**
- **Team Name – InfineX**

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Smart Low-Visibility Detection and Collision Prevention System for Mine Vehicles

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CHALLENGE WE'RE SOLVING

Low visibility conditions such as fog, dust, smoke and blind curves in mines lead to frequent accidents and vehicle collisions, threatening the safety of miners and equipment.

OUR PROPOSED SOLUTION

A 360-degree vehicle safety system that uses ultrasonic sensing, vehicle-to-vehicle (V2V) communication and predictive risk analysis.

1. **360° Detection & Monitoring**
   - Constantly monitors front, rear and side areas to locate nearby vehicles / obstacles in fog, dust and low visibility conditions.
2. **Risk Prediction Engine**
   - Combines distance, position, speed and direction to predict potential collision risk.
3. **360° Threat Map**
   - Provides a real-time 360° threat map, indicating the direction and degree of risk posed by the approaching vehicle.
4. **Intelligent Alerts**
   - Instant multilingual voice and visual alerts to the driver; critical alerts to the control-room.

ADDRESSING THE CORE PROBLEM

1. **360° Low-Visibility Detection:** Continuously detects vehicles/obstacles in all directions even in fog, dust and low visibility situations.
2. **Predictive Collision Risk Analysis:** Uses relative motion parameters to predict possible collisions before they occur.
3. **Real-time Threat Mapping:** Generates a 360° threat map showing the direction and severity of risk.
4. **Timely Alerts & Notifications:** Provides instant multilingual voice and visual warnings to drivers and critical alerts to the control-room.

HOW IT ADDRESSES THE PROBLEM

- Detects vehicles approaching even when visibility is reduced by fog, dust or blind curves.
- Gives early warnings of a possible crash, without relying solely on sight.
- Ranks multiple threats according to predicted collision risk.
- Directional mapping and voice alerts help drivers to understand where the danger is coming from.
- Modular retrofit for existing mine vehicles.

INNOVATION & UNIQUENESS

1. **Cooperative Safety**
   Vehicles share their position, speed and direction using V2V communications.
2. **Predictive, Not Reactive**
   Assesses risk using relative motion and Time-to-Collision (TTC) to predict potential collisions.
3. **360° Situational Awareness**
   Identifies and prioritizes hazards from all directions.
4. **Visibility-Aware Safety Mode**
   Increases warning sensitivity in poor visibility conditions and blind zones.
5. **Human-Centered Alerts**
   Provides direction-specific alerts in the driver's preferred language.
6. **Low-Cost Modular Architecture**
   Easy to install and retrofit on existing mine vehicles for scalable and affordable deployment.

IMPACT: | Improved Miner Safety | Reduced Accidents | Minimized Downtime | Increased Operational Efficiency | Scalable for All Mine Fleets

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TECHNICAL APPROACH

SOFTWARE

- C/C++ – ESP32 programming and sensor control
- Python – Sensor fusion & Time-to-Collision (TTC) risk calculation
- Python GUI – Vehicle position, radar range & risk display
- CSV/File Storage – Event and alert records
- Pre-recorded audio – Multilingual voice warnings

HARDWARE

- ESP32 – Vehicle safety controller
- mmWave Radar (24/77 GHz FMCW) – Long-range (20–40m), penetrates fog/dust/darkness — fixes the short range of ultrasonic-only sensing
- Ultrasonic Sensors – Short-range (<5m) close-proximity backup
- DGPS Module – Sub-metre accurate location (upgraded from plain GPS for reliable close-range V2V positioning)
- Wireless Comm. (RF/LoRa) – Vehicle-to-vehicle data exchange
- Buzzer + Speaker – Emergency alert

WORKING FLOW

1. Sense
Radar (20–40m, fog-penetrating) + Ultrasonic (<5m) + Vehicle Data → Detect nearby vehicles/obstacles beyond visual range, even in dense fog.

2. Communicate
V2V Communication → Share vehicle ID, DGPS position, speed & direction between nearby vehicles.

3. Fuse Data
Edge Controller combines radar + ultrasonic readings, V2V data and visibility condition (sensor fusion reduces false alarms from dust/rain).

4. Predict Risk
Collision Prediction Engine calculates relative motion + Time-to-Collision (TTC) and identifies the most dangerous approaching vehicle.

5. Alert Driver
360° Threat + multilingual voice alert; shows direction, distance and risk level; gives immediate instruction.

6. Monitor
Mine Control Dashboard sends critical alerts and vehicle status to the control room.

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FEASIBILITY AND VIABILITY

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1 Feasibility of the Idea

The proposed system is technically and practically feasible as it combines technologies such as cameras, LiDAR/radar, proximity sensors, GPS/GNSS and AI-based object detection. These sensors can continuously monitor the surroundings of mine vehicles and identify workers, vehicles, machinery and obstacles, even when dust, fog, darkness or other conditions reduce visibility. Since the system can be designed as an add-on module, it can potentially be integrated with existing mine vehicles without completely replacing their current systems.

2 Potential Challenges and Risks

Mining environments are extremely harsh, with dust, vibration, moisture, uneven terrain and changing visibility potentially affecting sensor performance. Individual sensors may sometimes provide inaccurate readings, resulting in false or missed detections. There is also the challenge of initial installation cost, maintenance and limited network connectivity, particularly in remote or underground mining areas. Excessive warnings could also distract operators or lead to reduced trust in the system.

3 Strategies for Overcoming Challenges

These challenges can be addressed through multi-sensor fusion, where information from cameras, LiDAR/radar and proximity sensors is combined rather than depending on one technology. AI-based filtering can distinguish genuine hazards from irrelevant objects and provide risk-based alerts according to the severity of the situation. Edge computing can allow critical decisions to be processed directly within the vehicle, ensuring faster response even without continuous internet connectivity. Rugged hardware, regular calibration and phased field testing can further improve reliability.

4 Overall Viability

The solution is viable because it provides an additional safety layer without fundamentally changing existing mining operations. Apart from improving safety during low-visibility conditions, the same system can eventually support collision prevention, vehicle tracking, traffic management and real-time safety monitoring. This makes the solution scalable and useful for both surface and underground mining environments.

KEY BENEFITS: Enhances Miner Safety | Reduces Accidents | Minimizes Downtime | Scalable & Cost-Effective | Useful for Surface & Underground Mining

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IMPACT AND BENEFITS

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SOCIAL	ECONOMIC	ENVIRONMENTAL	SCALABILITY
• Reduces the risk of vehicle-vehicle and vehicle-obstacle collisions. • Provides direction-specific multilingual voice alerts. • Improves safety for drivers, workers and other mine personnel.	• Reduces accident-related downtime, vehicle damage and operational losses. • Supports safer and more continuous vehicle movement. • Modular retrofit design reduces the need to replace existing vehicles.	• Better vehicle coordination can reduce unnecessary idling and movement. • Supports efficient mine operations and reduced avoidable fuel consumption. • Safety data can support better operational planning.	• Can scale from one vehicle to multiple vehicles. • Can expand into a mine-wide connected safety network. • Historical safety data can identify recurring high-risk zones.

"From individual vehicle safety to connected mine-wide protection."

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RESEARCH AND REFERENCES

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- **ILO – Code of Practice: Safety and Health in Opencast Mines** : https://www.ilo.org/resource/other/code-practice-safety-and-health-opencast-mines?utm_source
- **ILO – Safety and Health in Opencast Mines** https://www.ilo.org/sites/default/files/wcmsp5/groups/public/%40ed_dialogue/%40sector/documents/normativeinstrument/wcms_617123.pdf?utm_source
- **ILO – Mining Safety Resources**
- https://www.ilo.org/topics-and-sectors/industries-and-sectors/mining?utm_source